Symbol	Parameter	Conditions		Min	Typ	Max	Unit
I_{DDA}(SCALER)	Scaler static consumption from V_{DDA}	BRG_EN=0 (bridge disable)		-	0.2	0.3	µA
		BRG_EN=1 (bridge enable)		-	0.82	1	
t_{START_SCALER}	Scaler startup time	-		-	140	250	µs
t_{START}	Comparator startup time to reach propagation delay specification	High-speed mode		-	2	5	µs
		Medium mode		-	5	20	
t_D[2]	Propagation delay for 200 mV step with 100 mV overdrive	High-speed mode		-	50	80	ns
		Medium mode		-	0.5	0.9	µs
	Propagation delay for step > 200 mV with 100 mV overdrive only on positive inputs	High-speed mode		-	50	120	ns
		Medium mode		-	0.5	1.2	µs
V_{offset}	Comparator offset error	Full common mode range		-	-	±20	mV
V_{hys}	Comparator hysteresis	No hysteresis		-	0	-	mV
		Low hysteresis		-	10	-	
		Medium hysteresis		-	20	-	
		High hysteresis		-	30	-	
I_{DDA}(COMP)	Comparator consumption from V_{DDA}	Medium mode	Static	-	5	9	µA
			With 50 kHz ±100 mV overdrive square signal	-	6	-	
		High-speed mode	Static	-	70	110	
			With 50 kHz ±100 mV overdrive square signal	-	75	-	

1. *Refer to Section 5.3.5: Embedded voltage reference.*
2. *Evaluated by characterization - Not tested in production.*

5.3.23 Timer characteristics

The parameters given in Section 5.3.23, Table 64, and Section 5.3.23 are specified by design, not tested in production.

Refer to Table 46. I/O static characteristics for details on the input/output alternate function characteristics (output compare, input capture, external clock, PWM output).

Table 63. TIMx characteristics

Symbol	Parameter	Conditions	Min	Max	Unit[1]
$t_{res(TIM)}$	Timer resolution time	-	1	-	$t_{TIMxCLK}$
		$f_{TIMxCLK}$ = 144 MHz	6.9	-	ns
fEXT	Timer external clock frequency on CH1 to CH4	-	0	$f_{TIMxCLK}/2$	MHz
		$f_{TIMxCLK}$ = 144 MHz	0	72	
ResTIM	Timer resolution	TIMx (except TIM2/TIM5)	-	16	bit
		TIM2/TIM5	-	32	
$t_{COUNTER}$	16-bit counter clock period	-	1	65536	$t_{TIMxCLK}$
		$f_{TIMxCLK}$ = 144 MHz	0.007	455.1	µs
t_{MAX_COUNT}	Maximum possible count with 32-bit counter	-	-	4, 294, 967, 296	$t_{TIMxCLK}$
		$f_{TIMxCLK}$ = 144 MHz	-	29.826	s

1. *TIMx, is used as a general term in which x stands for 1, 2, 5, 6, 7, 8, 12, 15, 16, 17.*